MissionMind — Revised Research Proposal

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Draft for supervisor review. Every result in this document is a hypothesis to be tested; no experiment has yet been run. Companion document: MissionMind — Project Audit & Research Direction.

1. Proposed title

MissionMind: Evidence-Anchored Causal-Factor Graphs for Cross-Mission Analogy Retrieval over Spaceflight Mishap Investigation Reports

Short title: *Evidence-Anchored Graph RAG for Spaceflight Mishap Analysis*. The title names the task (cross-mission analogy retrieval), the mechanism (evidence-anchored causal-factor graph) and the data (mishap investigation reports). It drops "hallucination-free" and "cross-agency causal discovery", which the evidence cannot support.

2. Abstract

Spaceflight mishap investigation boards publish detailed reports. Their lessons are scattered across long, unstructured documents from different agencies and decades, so causal patterns shared between missions are found only when an analyst happens to read both reports. Retrieval-augmented generation (RAG) and graph-based RAG have been applied to aviation, maritime, automotive and manufacturing failure records. Those collections are large and semi-structured. They do not represent the spaceflight setting of few, very long, heterogeneous reports, and none evaluates cross-case causal analogy against human judgement.

We propose MissionMind, a framework that (i) recovers section-level fields and page provenance from mishap reports; (ii) extracts causal factors into a fixed taxonomy, accepting a graph edge only when its supporting source span is verified; and (iii) fuses lexical, dense and causal-graph retrieval to answer single-mission, cross-mission and analogy questions with page-level citations. We will also build a human-annotated benchmark from 25–35 public reports: causal-factor annotations, graded cross-mission analogy judgements, and a categorised question set including unanswerable questions, with inter-annotator agreement reported.

MissionMind will be compared with LLM-only, BM25, dense, multi-field hybrid (the method of our 2026 IEEE base paper), the current MissionMind prototype and a published GraphRAG system. Ablations and paired significance tests will isolate the

contribution of each component. *No results are reported here; the hypotheses and acceptance criteria are stated in Sections 14 and 16.*

3. Introduction and research motivation

The engineering value of a mishap report lies in its causal findings: what failed, why, and what organisational conditions allowed it. Well-known losses illustrate how differently those findings are recorded. Ariane 5 Flight 501 (1996) was lost to an unprotected data conversion in reused inertial-reference software. Mars Climate Orbiter (1999) was lost to a metric/imperial unit mismatch in a ground software file. Both inquiry boards also pointed to verification and interface-management weaknesses, but the two reports share almost no vocabulary.

This is the core difficulty. Causally analogous failures are written in different words, at different lengths (the Ariane 501 report is about 6,000 words; the Columbia CAIB Volume I about 175,000), with different section structures and by different agencies. Keyword search misses the analogy. Dense retrieval over fixed-size chunks retrieves topically similar text, not causally similar failures. A plain LLM may "remember" famous accidents but cannot cite them, and it can confabulate a plausible link.

Graph-based RAG promises to make shared structure explicit. Recent work, however, warns that graphs built by unverified LLM extraction propagate hallucinations (AGRAG, IEEE ICDE 2026). The closest 2026 IEEE work, multi-field hybrid RAG over 13,329 maritime tribunal reports (Kim & Kim, IEEE CASE 2026), relies on standard report fields and metadata-derived proxy labels. It names graph-based reasoning as future work. Spaceflight reports have neither standard fields nor large volumes, so it is an open question whether, and for which question types, a verified causal graph adds value over strong hybrid retrieval. MissionMind is designed to answer that question with evidence rather than to assume it.

4. Problem statement

Given a small corpus D of long, unstructured, multi-agency spaceflight mishap reports and an analyst query q, the system must do two things:

1. Retrieve the evidence passages, and for analogy queries the ranked set of missions, that are *causally* relevant to q.
2. Generate an answer in which every factual claim cites a page in D, or abstain when D does not support an answer.

Three properties make this hard, and they define what a solution must demonstrate:

- **Lexical divergence.** Causally equivalent factors are described in unrelated vocabulary across agencies and eras.
- **Structural heterogeneity.** There are no shared fields, so the field-aware indexing that works for tribunal records must be recovered from each report's own structure.
- **Verifiability.** In a safety context an uncited or fabricated causal link is worse than no answer. Graph facts and answer claims must be traceable to source spans.

The current prototype addresses none of these demonstrably: its query layer is broken, its graph would be 99.4% unverified LLM output, and nothing has been evaluated (see the audit document).

5. Research gap

The gap is the combination of four things no reviewed work does together: spaceflight mishap reports, verified causal graph construction, cross-case analogy retrieval, and human-validated evaluation.

Gap	Evidence from the literature	What MissionMind does
G1. Graph reasoning over precedents is untested in RCA retrieval	Kim & Kim [1] list "graph-based reasoning" as future work; their system is flat hybrid retrieval	Adds a typed causal-factor graph channel to the base paper's hybrid retrieval and measures its marginal value
G2. Evaluation rests on proxies	[1] uses metadata proxy relevance, rule-based tags and an unvalidated LLM judge, and lists all three as limitations	Human graded relevance, human causal-factor labels, judge-vs-human agreement; meta-evaluation of the proxy
G3. LLM-built graphs are unverified	AGRAG [3] identifies hallucinated LLM entity extraction as a core GraphRAG weakness; [2] reports extraction F1 but no per-edge provenance in its abstract	Each edge carries a source quote and page and is kept only if a deterministic span check and an entailment check pass
G4. Long, unstructured, few-document regime	Prior failure-record RAG [1], [2], [5], [6] uses thousands of short or semi-structured records	25–35 reports of 6k–175k words; field structure recovered by section segmentation

Gap	Evidence from the literature	What MissionMind does
G5. Cross-mission analogy is analysed manually	Studies such as Tafazoli [24] classify failures by hand; no retrieval benchmark exists	Defines and evaluates mission-level causal analogy retrieval

Scope honesty: G4 and G5 are claimed only for the corpus actually built. Generalisation to other domains is future work (Section 18).

6. Objectives and research questions

Objectives

1. **O1 Corpus.** Build a reproducible corpus of 25–35 public spaceflight mishap reports with a manifest (source URL, version, SHA-256), page provenance and recovered section fields.
2. **O2 Benchmark.** Create human annotations: causal factors on about 350 passages, graded analogy relevance for all mission pairs, and about 150 categorised questions with evidence pages. Report inter-annotator agreement.
3. **O3 Graph.** Construct an evidence-anchored causal-factor graph over a fixed taxonomy, with per-edge source verification.
4. **O4 Retrieval and generation.** Implement graph-augmented hybrid retrieval and citation-constrained generation with abstention.
5. **O5 Evaluation.** Compare against six baselines with ablations, paired significance tests and judge validation.

Research questions

ID	Question	Primary evidence
RQ1	Does span-and-entailment verification raise the precision of LLM causal-factor extraction, and at what cost in recall?	Extraction P/R/F1 vs the D1 gold, with and without verification
RQ2	Does a verified causal-factor graph improve cross-mission analogy retrieval over the base paper's multi-field hybrid retrieval?	Mission-level nDCG@5 on D2; P vs B3
RQ3	For which question categories does the graph help, hurt, or make no difference?	Per-category answer and citation metrics on D3

ID	Question	Primary evidence
RQ4	How well do cheap proxies (metadata relevance, LLM judge) agree with human judgement in this domain?	Spearman ρ / Cohen's κ between proxy and human labels

RQ4 directly tests the base paper's evaluation assumptions. A negative answer to RQ2 or RQ3 is still a reportable finding.

7. Base paper and justification

Base paper [1]: S. Kim and S. Kim, "Multi-Field Hybrid Retrieval-Augmented Generation for Maritime Accident Root Cause Analysis," *2026 IEEE 22nd Int. Conf. Automation Science and Engineering (CASE)*, pp. 238–243, Aug. 2026. Verified through the IEEE DOI and Crossref metadata; full text read via arXiv 2606.13249.

Primary supporting paper [2]: W. Sanyaolu and D. P. Josyula, IEEE ICSC 2026, a schema-guided LLM knowledge graph from 6,193 NTSB reports. Only the abstract was accessible.

Why this base paper

- **Same task family.** Precedent retrieval and grounded RCA drafting from official accident investigation reports.
- **Explicit future work and limitations we can act on.** Graph-based reasoning (G1) and proxy-based evaluation (G2).
- **A concrete, reimplementable method.** Per-field BM25 + bge-m3 dense retrieval fused by RRF (k = 60). This becomes our strongest non-graph baseline (B3), so any graph gain is measured against the base paper's own approach rather than a weak strawman.
- **Verifiable and open.** IEEE-published in 2026, with a CC BY 4.0 full text.

What we do not inherit. Its corpus scale (13k reports) is not available in our domain. Its metadata proxy is not reused as ground truth; it is re-examined in RQ4. Its 80B generator is replaced by open 7–8B models that fit an 8 GB GPU, which is a deliberate reproducibility choice and a stated limitation.

8. Existing system and its limitations

The existing system has two parts: the published state of the art (the base paper's flat hybrid RAG) and the current MissionMind prototype. Neither meets the requirements in Section 4.

Component	Current MissionMind (commit `5d0219e`)	Limitation	Disposition
Corpus	5 failure documents + 2 ISRO annual reports	Too few missions; wrong ESA source; no ISRO failure report	Rebuild (Stage 2)
Parsing	PyMuPDF + OCR fallback; ASCII stripping; whitespace collapse	Ligature loss, structure and page loss, OCR noise undetected	Keep extractor; replace cleaning
Chunking	250-word windows, 50 overlap	Cuts across sections; no fields	Replace with section-aware passages
Extraction	Case-sensitive gazetteer (gate passes 0.6% of chunks) → Llama-3 fallback	Effectively unverified LLM output; first-match root-cause labels	Replace with schema-constrained, verified extraction; keep the gazetteer as a rules baseline
Graph	`MERGE` by raw string; MiniLM similarity edges at an uncalibrated 0.65	Spurious cross-mission links from template names	Replace with a canonicalised taxonomy-typed graph
Retrieval	Vector top-3 + attached graph facts	Retriever class missing; API and CLI broken	Rebuild
Generation	Unconstrained prompt, no citations	Uncitable answers	Citation-constrained with abstention
Evaluation	Harness broken; placeholder data; same judge as generator	No evidence	Rebuild per Section 14

The base paper's own limitations, which MissionMind must not repeat, are summarised in Section 7 and analysed in the audit document.

9. Proposed methodology

The method has six modules (M1–M6). Each is labelled **[Established]** (used as published), **[Adapted]** (a published method changed for this setting) or **[Proposed]** (our contribution, to be validated).

M1. Corpus construction with provenance — [Established practice]

Collect 25–35 public reports into a manifest recording agency, mission, year, source URL, document version, SHA-256 and licence. ISRO annual reports are kept only as *distractor* documents, to test whether retrieval avoids non-failure text. Candidate list and selection rules: Section 12.

M2. Structure-aware parsing and field recovery — [Adapted from [1]]

- NFKC Unicode normalisation **before** any filtering (fixes ligature loss). Keep page numbers and character offsets for every passage.
- Heading detection from PyMuPDF font size and weight plus numbering patterns. Passages = paragraphs merged up to about 200–300 tokens, never crossing a section boundary.
- Each section is assigned a pseudo-field (*Narrative*, *Analysis/Findings*, *Causes*, *Recommendations*, *Other*) by heading rules, with an LLM classifier for unmatched headings. This recovers the base paper's multi-field structure where reports have no fixed fields. Accuracy is validated on a 100-section sample.
- OCR quality is measured, not assumed. Character error rate is estimated on about 20 hand-transcribed sampled pages, and pages over threshold are re-OCR'd or flagged.

M3. Causal schema and taxonomy — [Adapted]

Nodes: `Mission`, `Event` (the failure), `Subsystem` and `Phase` (controlled vocabularies), `CausalFactor` (typed by a two-level taxonomy), `Recommendation`, `Passage`. Edges: `(CausalFactor)-[CONTRIBUTED_TO {role}]->(Event)`, where role is root, contributing or proximate, *as stated by the board*; `(Recommendation)-[ADDRESSES]->(CausalFactor)`; `(edge)-[EVIDENCED_BY]->(Passage)`.

The taxonomy has L1 classes such as technical design, hardware/material, software, environment, operations/human, organisational/process, verification and validation, and interface/communication. Below them sit about 20–30 L2 factors. It is adapted from established safety taxonomies (HFACS [27] for human and organisational tiers; Leveson's STAMP control-flaw view [28]) and subsystem categories used in failure studies [24]. It is **frozen before test labels are seen**, and the final version needs supervisor sign-off.

M4. Evidence-anchored extraction — [Proposed]

1. An open 7–8B LLM extracts JSON records from each *Analysis/Causes* section, with JSON-schema-constrained decoding at temperature 0. Each record holds factor text, L2 label, role, subsystem, phase, and a **verbatim supporting quote**.

2. **Span check (deterministic).** The quote must match the source passage at normalised similarity ⩾ 0.9. The page and offsets are attached, and records with no match are rejected.
3. **Entailment check.** An NLI cross-encoder must judge that the quote entails a templated statement of the record (e.g. "X contributed to the loss of M"). The threshold is tuned on dev data only.
4. **Canonicalisation.** Factors are mapped to L2 by embedding nearest-neighbour plus LLM tie-break, and subsystems are linked to the controlled list. Nodes are merged by canonical ID, never by raw string. This removes the template-name collisions found in the audit.

The current gazetteer NER, fixed to be case-insensitive, is kept as a rules baseline. Unverified LLM extraction is kept as an ablation.

M5. Graph-augmented hybrid retrieval — [Adapted from [1] + Proposed graph channel]

Five rankings are fused by RRF [13] with k = 60, tuned on dev: BM25 and dense (bge-m3 [15]) over *Causes* fields, the same two over all fields, and one graph ranking. The graph channel links query mentions to missions, factors and subsystems. It then expands to passages that evidence factors shared with the linked missions, weighting each factor by its rarity across missions:

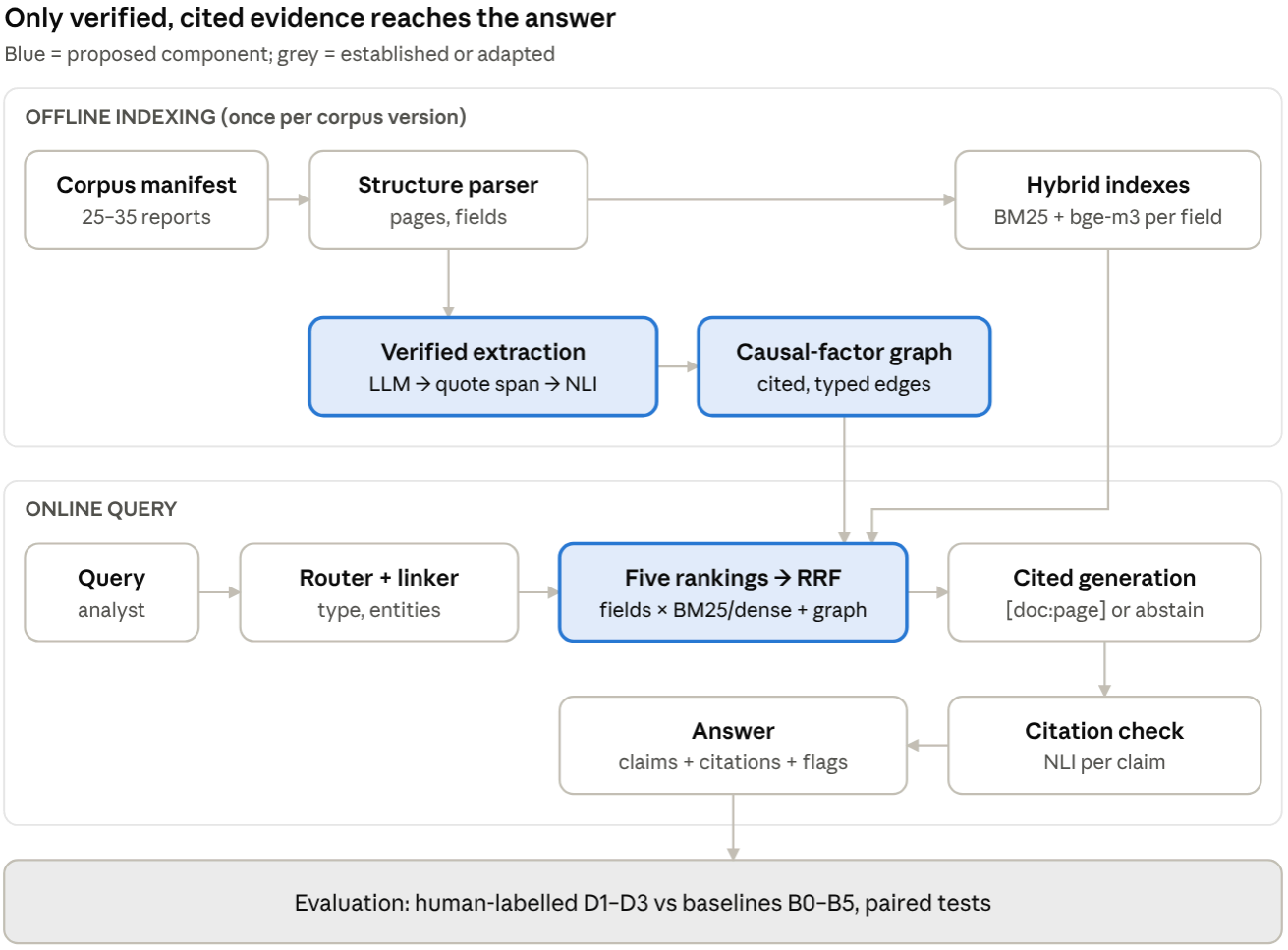
```
\mathrm{RRF}(p)=\sum_{r\in R}\frac{1}{k+\mathrm{rank}_r(p)},\qquad
\mathrm{analogy}(m_i,m_j)=\frac{\sum_{f\in F_i\cap F_j}\mathrm{idf}(f)}
{\sum_{f\in F_i\cup F_j}\mathrm{idf}(f)}
```

Here F_i is the set of verified L2 factors of mission m_i, and idf(f) = log(N / n_f) over N missions. Analogy queries rank missions by a combination of this weighted Jaccard score and the best passage-level dense similarity. The weight is tuned on dev missions only.

M6. Citation-constrained generation with abstention — [Adapted]

The same generator, prompt template and context budget (e.g. 8 passages) are used for every system. The prompt requires a `[doc:page]` citation on every claim and an explicit "not supported by the corpus" answer when the evidence is insufficient. Citations are checked after generation: does the cited passage exist, and does it entail the claim?

10. Improved model architecture and framework



MissionMind proposed architecture · offline indexing, online query, evaluation

The graph enters the answer only through the fusion step and only as cited passages, so a graph error can lower rank quality but cannot inject an uncited claim. Blue components are the ones whose value the experiments must prove (RQ1–RQ3); removing each one is an ablation in Section 13.

11. Detailed workflow and component interactions

The system runs in two phases. An offline indexing phase writes every artefact to versioned files and Neo4j. An online query phase only reads them, so experiments are repeatable from a frozen index.

Offline indexing (run once per corpus version)

1. `manifest.yaml` → fetch and checksum PDFs. A run stops if any checksum differs from the manifest.

2. Parser → `passages.jsonl` (passage id, doc, page, offsets, section path, pseudo-field, text). An OCR quality report is written alongside.
3. Indexers → BM25 index and bge-m3 vectors per field (Neo4j vector index or FAISS; both read the same `passages.jsonl`).
4. Extractor → `candidates.jsonl` (raw LLM records, prompt hash, model tag) → verifier → `verified.jsonl` plus `rejected.jsonl` with the rejection reason.
5. Canonicaliser → graph load: `UNWIND` batch writes of Mission, Event, CausalFactor, Subsystem, Phase, Recommendation and Passage nodes. Every edge carries its evidence passage id.
6. Analogy pre-computation → mission × mission factor-overlap matrix.

Online query

1. Query router (rules + LLM): assigns a type — single-mission factual or causal, cross-mission comparison, analogy/aggregation, or out-of-scope.
2. Entity linker: maps mentions to missions, factors and subsystems in the graph.
3. Retrieval: five rankings computed in parallel → RRF fusion → top-k passages. For analogy queries, a ranked mission list is also returned.
4. Context builder: passages are ordered by fused score and labelled `[doc:page | field]`. For graph hits, a one-line fact ("factor F shared by M1, M2; evidence p. x, y") is added, always paired with its passage.
5. Generator → answer with citations, or an abstention.
6. Citation verifier: each claim's citation is checked by NLI. Unsupported claims are flagged in the API response.
7. Logger: query, router type, retrieved ids, prompt hash, model tag, answer and verification flags go to `runs/<run_id>/`, which the evaluation reads directly.

Where the current code is reused: PyMuPDF and OCR fallback (step 2), the Neo4j connector (step 5), the FastAPI shell (online steps), and the metric functions once re-tested.

12. Dataset, tools, technologies and experimental requirements

12.1 Corpus

Inclusion rules. An official investigation, inquiry-board or failure-analysis document (full report or official public summary) for a spaceflight loss or major anomaly; publicly accessible; English or official English translation; causal findings stated. Journal articles about a mission are excluded as primary sources.

Already on disk and valid (4): Challenger (Rogers Commission Vol. 1, 1986), Columbia (CAIB Vol. I, 2003), Mars Climate Orbiter (MIB Phase I, 1999), Ariane 5 Flight 501 (Inquiry Board, 1996).

To replace (1): ExoMars Schiaparelli → ESA *Schiaparelli Anomaly Inquiry* (DG-I/2017/546/TTN, 2017).

Candidates to verify in Stage 2. Availability and licence are *not yet checked* for any of these. The list is a search plan, not a claim:

- NASA: Mars Polar Lander (JPL Special Review Board, 2000), Genesis MIB, CONTOUR MIB, DART MIB, WIRE, NOAA-N Prime, Orbiting Carbon Observatory, Glory, Mars Observer, Lewis, Apollo 13 Review Board, Apollo 204 Review Board
- ESA and partners: SOHO mission interruption (ESA/NASA, 1998), Beagle 2 Commission of Inquiry, Vega VV15 and VV17 independent inquiry summaries
- JAXA: Hitomi (ASTRO-H) anomaly investigation report
- ISRO: GSLV-F10 Failure Analysis Committee public statement; other official public statements where available
- Distractors: the two ISRO annual reports already on disk

Target: 25–35 missions, at least 3 agencies. Expected reality: NASA-heavy. The agency distribution will be reported, and no cross-agency claim will exceed it.

12.2 Human-annotated benchmark (planned; none exists yet)

Set	Content	Size (target)	Second annotator
D1 Extraction gold	Passages stratified by pseudo-field and mission, labelled with L2 factors, role, subsystem, phase	~350 passages	⩾ 25%
D2 Analogy relevance	Every mission pair graded 0/1/2 for shared causal mechanism, from 1-page mission cards	300–600 pairs (25–35 missions)	⩾ 25%
D3 QA set	~150 questions: single-mission factual (30), single-mission causal (30), cross-mission comparison (30), analogy/aggregation (40), unanswerable (20); each with key points and evidence pages	~150	Review of all; ⩾ 25% re-answered

Dev/test split: about 30% dev (tuning, threshold selection) and 70% test (touched once). Analogy tuning uses dev missions only. Annotation guidelines are written before labelling and revised at most once after the pilot round.

12.3 Tools

Layer	Tool (open, local)	Status
PDF / OCR	PyMuPDF, Tesseract via pytesseract	Reused
Lexical retrieval	BM25 (`rank_bm25` or Pyserini)	New
Dense retrieval	bge-m3 [15]; MiniLM [16] and E5 [31] as ablations	New
Graph store	Neo4j 5 Community (Docker)	Reused
LLMs	Llama-3.1-8B-Instruct and Qwen2.5-7B-Instruct, 4-bit, via Ollama, JSON-schema output	Partly reused
Verification	NLI cross-encoder (DeBERTa-v3 class; exact model chosen on dev)	New
GraphRAG baseline	LightRAG [8] (primary); Microsoft GraphRAG [7] on a subset if compute allows	New
Evaluation	Own metric code + RAGAS [17] for comparison; SciPy for statistics	Partly reused
Annotation	Label Studio or a spreadsheet with a frozen schema	New
Reproducibility	`pyproject.toml` + lock file, YAML configs, fixed seeds, run logs, pytest	New

13. Baselines and comparative evaluation strategy

The key comparison is P against B3, the base paper's method reimplemented on our corpus. Every other baseline exists to locate where gains or losses come from.

ID	System	Purpose
B0	LLM only, no retrieval	Measures parametric knowledge; contamination check on famous accidents
B1	BM25 over passages	Lexical floor

ID	System	Purpose
B2	Dense (bge-m3) over passages	Standard vector RAG
B3	Multi-field BM25 + dense with RRF [1]	**Base paper method** (strongest non-graph baseline)
B4	Current MissionMind (repaired: vector top-k + unverified graph facts)	Shows what the redesign changes
B5	LightRAG [8] (Microsoft GraphRAG [7] on a subset if feasible)	Published GraphRAG reference
P	B3 + verified causal-factor graph channel + citation-constrained generation	Proposed

Extraction comparisons (RQ1): case-insensitive gazetteer NER · LLM unverified · LLM + span check · LLM + span + NLI (P).

Ablations of P: − graph channel (= B3 + M6) · − verification (unverified graph) · − canonicalisation (raw-string nodes) · − fields (all text as one field) · graph channel only · idf weighting off · embedder swap (MiniLM / E5 / bge-m3) · generator swap (Llama-3.1-8B / Qwen2.5-7B).

Fairness controls:

- Identical corpus version, passages, generator, decoding settings, prompt template and context budget across systems. Only the retrieved content differs.
- Hyperparameters (RRF k, thresholds, analogy weight) are tuned on dev for *every* system with the same search budget, not just for P.
- B5 is run with its documented default settings plus the same generator. Any deviation is reported.
- Test questions are written from reports, before any system output is seen.

Contamination check. B0 accuracy is reported per mission. If B0 does well on famous missions, results are also reported on the less-documented subset, and unanswerable questions test whether systems answer from memory.

14. Evaluation metrics and validation methodology

Every headline metric is computed against human labels. Proxy metrics are reported only beside them, and their agreement with humans is itself measured (RQ4).

Level	Metric	Gold	RQ
Annotation reliability	Cohen's κ (categorical), Krippendorff's α (graded), % agreement	Double-labelled 25%	All
Parsing	Pseudo-field accuracy; OCR character error rate on sampled pages	100 sections; ~20 pages	—
Extraction	Micro and macro P/R/F1 at L1 and L2; role accuracy; evidence-span validity rate; % of edges rejected by each check	D1	RQ1
Passage retrieval	Recall@k (k = 5, 10, 20), nDCG@10, MRR	D3 evidence pages	RQ3
Analogy retrieval	Mission-level nDCG@5, P@3, MAP	D2 graded pairs	RQ2
Answer quality	Key-point recall (human-scored on test); claim-level support rate in the style of FActScore [19]	D3 key points + corpus	RQ3
Citations	Citation precision (cited passage entails claim); citation recall (claims with a valid citation)	NLI + human check on a sample	RQ3
Abstention	Correct-abstention rate on unanswerable items; false-abstention rate on answerable ones	D3 category 5	RQ3
Judge validity	Spearman ρ and κ between LLM judge and human ratings on ≥ 25% of answers; judge model from a different family than the generator	Human ratings	RQ4
Proxy validity	Spearman ρ between a metadata proxy (as in [1]) and human analogy grades	D2	RQ4
Cost	Indexing time, query latency, LLM calls, peak VRAM	Logs	—

Statistical protocol (fixed before the test set is opened)

- Primary comparison: P vs B3. Primary metrics: analogy nDCG@5 and key-point recall. Everything else is secondary.

- Paired bootstrap with 10,000 resamples over questions (or mission queries) for 95% CIs of differences [23]. A paired permutation test gives p-values, with Holm–Bonferroni correction across secondary metrics and categories.
- Effect sizes are reported alongside p-values. With about 105 test questions, small differences will not be detectable; a power note will be reported rather than over-claiming.
- Generation is repeated with 3 seeds (sampled decoding for robustness runs; temperature 0 for the main run), and the variance is reported.
- Every number in the paper is regenerated from logged runs by one script. No hand-copied results.

15. Proposed contributions and qualified novelty claims

There are four contributions. Each is phrased the way it may be claimed in a paper *if and only if* the evidence in the right-hand column materialises.

#	Contribution	Novelty level	Claim allowed only if
C1	A reproducible, provenance-preserving corpus and a human-annotated benchmark for spaceflight mishap analysis (causal factors, cross-mission analogy grades, categorised QA with unanswerables)	New resource in this domain; built with established annotation practice	⩾ 25 missions released or reconstructible from the manifest; agreement statistics reported, with κ ⩾ 0.6 targeted and lower values reported honestly
C2	Evidence-anchored causal-factor graph construction (schema-constrained LLM extraction + span and entailment verification + taxonomy canonicalisation)	Proposed combination of known parts; not a new algorithm	RQ1 shows a precision gain over unverified extraction with its recall cost quantified
C3	Graph-augmented multi-field hybrid retrieval, i.e. a graph channel fused into the base paper's method, with idf-weighted factor overlap for mission analogy	Adapted extension, answering future work stated in [1]	RQ2/RQ3 show a significant gain over B3 on at least one pre-registered primary metric or category. If not, the claim becomes a negative result

#	Contribution	Novelty level	Claim allowed only if
C4	An empirical study of when graphs help, and of proxy and LLM-judge validity, in a small-corpus safety domain	Empirical finding	Per-category analysis and RQ4 correlations reported with CIs

Will not be claimed: "hallucination-free", "deterministic", "automatic causal discovery", "cross-agency generality" beyond the agencies represented, or any superiority over systems not actually run.

16. Expected outcomes (hypotheses, not results)

Nothing below has been observed. Each item is a falsifiable hypothesis with the outcome that would refute it, so the paper can report either direction.

ID	Hypothesis	Rationale	Refuted if
H1	Verification raises extraction precision over unverified LLM extraction, at a measurable recall cost	Rejected records are those whose quote is absent or non-entailing	No significant precision difference on D1 test
H2	P outperforms B3 on mission-level analogy nDCG@5	Shared typed factors encode causal similarity that lexical and dense text similarity miss	95% CI of (P − B3) includes 0
H3	Graph gains concentrate in the cross-mission and analogy/aggregation categories, with little or no gain on single-mission factual questions	Single-mission facts sit in one passage that hybrid retrieval already finds	Gains are uniform or absent across categories
H4	Citation-constrained generation with abstention reduces unsupported claims relative to B2 and B4	The prompt and post-hoc check penalise uncited claims	Claim support rate not higher

ID	Hypothesis	Rationale	Refuted if
H5	The metadata proxy correlates only moderately with human analogy grades, and the LLM judge agrees imperfectly with humans	Proxies capture topical, not causal, similarity	Correlations are high, which would itself be useful evidence for [1]'s approach
H6	B0 scores well on famous missions and poorly on the rest	Pre-training exposure	No per-mission difference

Deliverables expected regardless of outcome: the corpus manifest and parsing code; annotation guidelines and labels (subject to source licences); a reproducible experiment pipeline; a paper reporting results with CIs; and a working demo (query → cited answer → evidence graph view).

17. Limitations, risks and mitigation strategies

The largest risks are corpus size and annotation reliability. Both limit what can be claimed rather than whether the project can finish.

Risk / limitation	Likelihood	Impact	Mitigation	Fallback
Fewer than 25 public reports with usable causal findings	Medium	High	Accept official public summaries; widen to major anomalies, not only losses	Report on the achieved N; frame as a pilot benchmark
No second annotator	Medium	High	Recruit a peer or the supervisor early (Stage 3 gate)	Self-agreement re-labelling after ≥ 2 weeks, clearly labelled as weaker evidence
Low inter-annotator agreement ($\kappa < 0.6$)	Medium	Medium	Pilot round, guideline revision, L1 fallback for unreliable L2 classes	Report at L1 only

Risk / limitation	Likelihood	Impact	Mitigation	Fallback
Verification discards too many true facts (recall collapse)	Medium	Medium	Tune thresholds on dev; report the precision–recall curve	Use span check only
Graph gives no gain over B3	Medium	Medium	Pre-registered per-category analysis	Publish as a negative or conditional finding (C4)
LLM pre-training contamination on famous accidents	High	Medium	B0 per-mission check; unanswerable set; less-known missions	Report results on the low-contamination subset
OCR noise in older scans (e.g. Challenger)	High	Medium	Measure CER; re-OCR at 300 dpi; flag pages	Exclude flagged pages and report it
8 GB GPU limits model size and speed	High	Low–Medium	4-bit 7–8B models; batch offline; cache	Smaller corpus slice for the GraphRAG baseline
Source licences restrict redistribution	Medium	Low	Release manifest + scripts, not PDFs, where needed	Release labels with offsets only
Agency imbalance (NASA-heavy; ISRO reports not public)	High	Medium	State it; no cross-agency generalisation claims	—
Small test set gives low statistical power	High	Medium	Paired tests, CIs, effect sizes; primary metrics fixed in advance	Report as directional evidence

Threats to validity are reported in the paper under internal (tuning leakage, single-site annotation), external (one domain, NASA-heavy) and construct (do analogy grades capture "causal similarity"?) headings.

18. Future-work directions

The base paper's two stated directions split cleanly. Graph-based reasoning is taken into scope now. Integrating formal regulations has no direct spaceflight analogue in this project, but standards-linked recommendations are a natural successor.

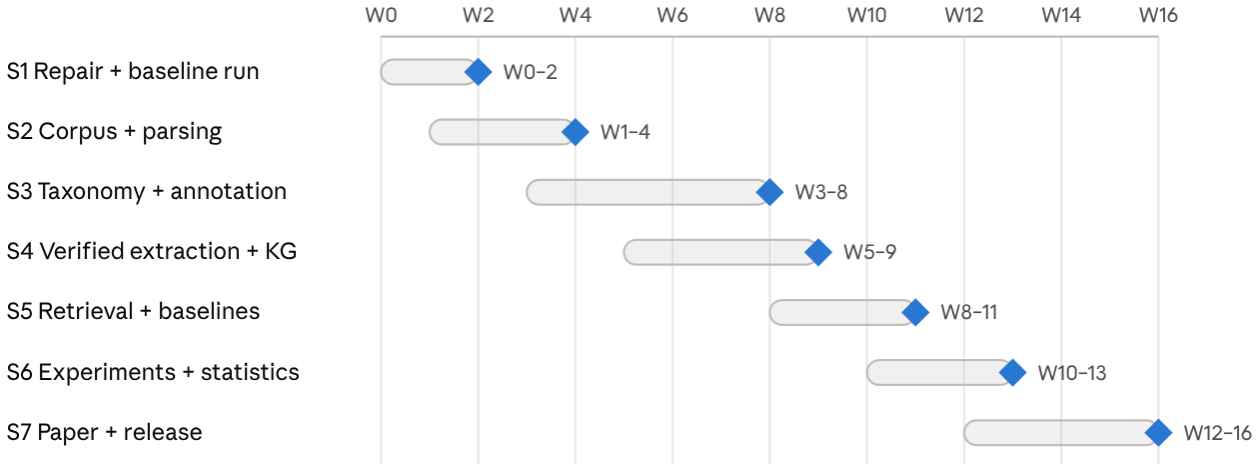
Direction	Source	Fit	Scope
Graph-based reasoning over precedents	[1] future work	Core of C3	**In scope**
Uncertainty-aware evaluation	[1] discussion	Abstention and selective accuracy only	**Partly in scope**
Better relationship extraction; hybrid retrieval	arXiv 2604.13101 (aviation GraphRAG preprint; venue unconfirmed)	Addressed by M4 and M5	**In scope**
Linking factors to standards and requirements (e.g. NASA or ECSS standards), the analogue of [1]'s COLREGs idea	[1] future work, adapted	Recommendation → standard edges	Future
Second domain (NTSB aviation or NASA lessons-learned records) to test generalisation	Reviewer expectation for a journal	Needed for an IEEE Access-level extension	Future
Fine-tuned domain extractor (e.g. continued pretraining of a SafeAeroBERT-style encoder [26])	[25], [26]	Requires more labels than D1	Future
STAMP/CAST control-structure generation from reports	[28]	Research-grade; hard to evaluate	Future
Multimodal evidence (figures, tables, telemetry plots)	2604.13101 limitation	High value, high cost	Future
User study with practising investigators	Construct validity	Needs institutional access	Future

Direction	Source	Fit	Scope
Robustness to corpus poisoning in shared-relation graphs	GraphRAG Under Fire [4]	Security angle for deployed systems	Future

19. Stage-wise implementation roadmap

Seven gated stages, about 16 weeks part-time

Planning estimate in weeks from approval; overlaps are deliberate



Blue diamond = gate: stage audit, then your explicit approval before the next stage starts

Stage plan · planning estimate, weeks after proposal approval

Only one stage is active at a time for decisions. Overlap is limited to preparation work, such as writing the annotation guideline during S2. Each gate produces a short stage report (what was done, acceptance check, unresolved issues), and the next stage starts only after your explicit approval.

Stage	Objective and key tasks	Deliverables	Acceptance criteria (checked at gate)	Main risk → corrective action
S1 Repair + baseline	Restore `GraphRAGRetriever`; fix `run_eval.py`; package (`pyproject`, lock file); env-only secrets; config + seeds; pytest; one end-to-end run on the current corpus; measure LLM throughput	Working pipeline; run log; measured B4 cost figures	One command rebuilds the index and answers a query; tests pass; run artefacts saved; throughput replaces the Section 20 estimates	Neo4j/Ollama environment issues → Docker compose file
S2 Corpus + parsing	Verify candidate reports; manifest with SHA-256; NFKC; page-aware, section-aware parsing; pseudo-fields; OCR CER sampling	`manifest.yaml`; `passages.jsonl`; parsing report	⩾ 25 missions (or a documented shortfall); pseudo-field accuracy ⩾ 90% on a 100-section sample; every passage has doc+page	Too few reports → include official summaries; reduce scope claim
S3 Taxonomy + annotation	Freeze schema/taxonomy (supervisor sign-off); write guidelines; pilot; label D1–D3; second annotator on ⩾ 25%	Guidelines; D1, D2, D3; agreement report; dev/test split	κ/α reported for all sets (target ⩾ 0.6); test set frozen and hashed before S4 tuning	Low agreement → one guideline revision; fall back to L1

Stage	Objective and key tasks	Deliverables	Acceptance criteria (checked at gate)	Main risk → corrective action
S4 Verified extraction + KG	Schema-constrained extraction; span check; NLI check; canonicaliser; graph load	`verified.jsonl`, `rejected.jsonl`; Neo4j graph; RQ1 dev results	Every edge has a resolvable evidence span; RQ1 measured on dev with all four extractor variants	Recall collapse → tune thresholds on dev; span-only variant
S5 Retrieval + baselines	Implement B0–B5 and P; router; citation-constrained generation; citation verifier; logging	All systems runnable from config; dev results	Fairness controls in Section 13 verified by config diff; all systems tuned on dev with equal budget	LightRAG cost → subset run, reported as such
S6 Experiments + statistics	Single test-set run; ablations; bootstrap/permutation tests; judge and proxy validation; error analysis	Results tables regenerated from logs; analysis notebook	Pre-registered primary comparisons reported with CIs, whatever the direction; no post-hoc metric switching	Null result → report as C4 finding

Stage	Objective and key tasks	Deliverables	Acceptance criteria (checked at gate)	Main risk → corrective action
S7 Paper + release	Venue choice; IEEE template draft; threats to validity; artefact packaging; demo	Paper draft; repository release; demo	Every number in the paper traced to a logged run; claims match Section 15 conditions; supervisor review	Page limit → move detail to an appendix or repository

20. Estimated computational resources and requirements

The whole study is planned to fit the existing laptop: an RTX 4060 Laptop GPU with 8 GB VRAM, CUDA available (verified). It needs no paid API, which also makes it reproducible. All figures below are **planning estimates, not measurements**. Stage 1 replaces them with measured throughput.

Item	Estimate	Basis (assumption)
Corpus size	~0.8–1.5 M words (25–35 reports)	Current 5 failure reports = 326k words; the remaining reports are mostly shorter
Passages	~4,000–7,000	200–300-token passages
Embedding (bge-m3, fp16)	< 1 h per corpus version	~2.3 GB VRAM; batched
LLM extraction (7–8B, 4-bit, ~5 GB VRAM)	~5–15 GPU-hours per full pass	Only Analysis/Causes sections (~30–40% of passages); ~200 output tokens each
NLI verification	< 1 h	Cross-encoder over candidate records
LightRAG / GraphRAG indexing	~5–20 GPU-hours, the largest uncertainty	LLM-heavy indexing; run on a subset if needed

Item	Estimate	Basis (assumption)
Answer generation	~7 systems × ~150 questions × 3 seeds ≈ 3,150 calls, ~5–10 h	Plus ablations of similar size
Judge (different family, e.g. Qwen2.5-7B if Llama generates)	~2–4 h	One call per answer
Storage	< 5 GB	PDFs ~0.3–0.5 GB; indices; logs
RAM	⩾ 16 GB recommended	Neo4j + models; not yet verified on this machine

Human effort (the real bottleneck)

Task	Estimate
Corpus search, verification, manifest	15–20 h
Mission cards (1 page each) for D2	15–20 h
D1 extraction gold (~350 passages × ~2 min)	~12 h + pilot
D2 analogy grades (300–600 pairs × ~1 min)	5–10 h
D3 questions, key points, evidence (~150 × ~10 min)	~25 h
Second annotator (25% of D1–D3)	~12–15 h of *their* time
Human answer scoring (test set, key points)	~15 h

21. References

Every entry below was checked on 2026-10-12 against a DOI resolver, Crossref, OpenAlex or arXiv. [7] and [31] are preprints. Entries [14] and [27] were not resolved by DOI and should be checked against the physical source before submission.

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22. Potential IEEE publication venues

A 6–8 page conference paper is the realistic first target. A journal follows only after scale-up or a second domain. Deadlines, page limits, indexing and fees change yearly; each must be confirmed on the official call for papers before choosing. None was verified for the 2027 cycle in this review.

Venue	Why it fits	Main risk	Readiness needed
IEEE Aerospace Conference (AERO; IEEE/AIAA, held annually at Big Sky, MT)	Spaceflight audience; values mission-assurance applications	Reviewers expect engineering relevance over NLP novelty, so the demo and case studies matter	Stages 1–6 complete
Annual Reliability and Maintainability Symposium (RAMS; IEEE Reliability Society co-sponsored)	Failure analysis, root cause and lessons learned are central topics	Less NLP methodology depth expected; must explain RAG accessibly	Stages 1–6
IEEE Int. Conf. on Semantic Computing (ICSC)	Venue of supporting paper [2]; KG + LLM fit	Shorter papers; competitive on method novelty	Stages 1–6
IEEE Int. Conf. on Automation Science and Engineering (CASE)	Venue of base paper [1]; direct dialogue with it	Automation focus; domain is adjacent rather than central	Stages 1–6
IEEE BigData (main track or workshops)	Venue of [5]; workshops accept applied KG/RAG studies	Main track wants scale we do not have; a workshop is the realistic fit	Stages 1–6

Venue	Why it fits	Main risk	Readiness needed
IEEE Access (journal, open access, article processing charge)	Broad scope; accepts thorough applied studies	Expects larger evaluation, generalisation and a complete artefact	Plus a second domain or a larger corpus
IEEE Trans. Reliability / *IEEE Trans. Aerospace and Electronic Systems*	High prestige in reliability / aerospace	High bar; needs substantial, validated results	Long-term only

Recommendation: aim the first paper at **AERO or RAMS** if the domain story and human-labelled benchmark are strongest. Aim at **ICSC** if the verification method (C2) shows the clearest gain. Decide after Stage 6 results, not before. No venue choice guarantees acceptance, and IEEE sponsorship alone does not establish research quality.